STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 2: Motion in a Straight Line

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define instantaneous velocity. [1]
2. Slope of an x-t graph? [1]
3. Acceleration at constant velocity? [1]
4. Can speed be negative? [1]
5. SI unit of acceleration? [1]
6. Runner returns to start: net displacement? [1]
7. Equation linking v,u,a,t? [1]
8. Area under v-t graph? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. 150 m in 30 s: average speed? [2]
10. Velocity 4 to 16 m/s in 6 s: acceleration? [2]
11. 10 m/s car brakes at 2 m/s^2: stopping time? [2]
12. Convert 90 km/h to m/s. [2]
13. Move 8 m east then 3 m west: distance/displacement? [2]
14. From rest at 3 m/s^2 for 4 s: final speed? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. u=6 m/s, a=2 m/s^2, t=5 s: find v,s. [3]
16. Train at 20 m/s stops in 5 s: a and distance? [3]
17. Velocity rises 2 to 14 m/s in 6 s: a and displacement? [3]
18. u=12 m/s, a=1.5 m/s^2 for 8 s: v,s? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Car at 25 m/s brakes at 5 m/s^2: stop time and distance? [4]
20. x=4+3t-t^2: v(t), a and time at rest? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Average speed = total distance/total time; average velocity = displacement/time.
- a = (v-u)/t; v = u + at; s = ut + (1/2)at^2; v^2 = u^2 + 2as.
- For constant acceleration, s = ((u+v)/2)t.
- At rest: v=0. Stopping distance under constant deceleration: s = u^2/(2|a|).
- Conversion: km/h to m/s, divide by 3.6; m/s to km/h, multiply by 3.6.
- Displacement = final position - initial position; distance is the full path length.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: v=dx/dt.
Final answer: v=dx/dt.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Velocity.
Final answer: Velocity.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: No.
Final answer: No.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: m s^-2.
Final answer: m s^-2.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: v=u+at.
Final answer: v=u+at.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Signed displacement.
Final answer: Signed displacement.

Question 9

Working: Average speed=total distance/time=150/30=5 m/s.
Final answer: 5 m/s.

Question 10

Working: a=(v-u)/t=(16-4)/6=12/6=2 m/s^2.
Final answer: 2 m/s^2.

Question 11

Working: Take final velocity v=0 and acceleration a=-2 m/s^2. Using v=u+at: 0=10-2t, so t=5 s.
Final answer: 5 s.

Question 12

Working: 90 km/h divided by 3.6 = 25 m/s.
Final answer: 25 m/s.

Question 13

Working: Distance=8+3=11 m. Taking east as positive, displacement=+8-3=+5 m, i.e. 5 m east.
Final answer: 11 m; 5 m east.

Question 14

Working: From rest u=0. Use v=u+at=0+3(4)=12 m/s.
Final answer: 12 m/s.

Question 15

Working: v=u+at=6+2(5)=16 m/s. Then s=ut+(1/2)at^2=6(5)+(1/2)(2)(25)=30+25=55 m.

Final answer: v=16 m/s; s=55 m.

Question 16

Working: a=(0-20)/5=-4 m/s^2. Distance=s=((u+v)/2)t=((20+0)/2)(5)=50 m.

Final answer: a=-4 m/s^2; s=50 m.

Question 17

Working: a=(14-2)/6=2 m/s^2. Displacement=average velocity x time=((2+14)/2)(6)=8(6)=48 m.

Final answer: a=2 m/s^2; displacement=48 m.

Question 18

Working: v=12+1.5(8)=24 m/s. s=ut+(1/2)at^2=12(8)+0.5(1.5)(64)=96+48=144 m.

Final answer: v=24 m/s; s=144 m.

Question 19

Working: Use v=u+at: 0=25-5t, hence t=5 s. Distance s=ut+(1/2)at^2=25(5)-0.5(5)(25)=125-62.5=62.5 m.

Final answer: 5 s; 62.5 m; acceleration opposes motion.

Question 20

Working: Given x=4+3t-t^2. Differentiate: v=dx/dt=3-2t; acceleration a=dv/dt=-2 m/s^2. At rest v=0, so 3-2t=0 and t=1.5 s.

Final answer: v=3-2t; a=-2 m/s^2; t=1.5 s.